

Installation Instructions

Transportation | Assembly | Operation | Maintenance



Product line:	Drycoolers
Series description:	V-Shape Vario Drycoolers
Series:	GFD

guntner.com/us

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1. Warranty statement

The supplier warrants the product to be free from defects in workmanship and materials under normal usage for the period outlined below. The supplier further warrants against defects in workmanship and materials of the product for a period of 24 months from the date of purchase (the "Warranty Period"), provided that the product is correctly installed and operated within the recommended limits of the manufacturer's technical documentation.

The supplier at its option shall repair or replace, free of charge to the purchaser, all components of the product which are or become defective during the Warranty Period as a result of defects in design, workmanship or materials, ordinary wear and tear excluded, provided, however, that:

- The product is applied correctly
- All operating and installation instructions for the product are complied with
- System component and piping design is in accordance with good refrigeration practice
- Nitrogen or an inert gas is introduced into the piping during the brazing of the piping installation

This warranty shall not include ordinary maintenance or cleaning of the product, or defects in the installation of the product.

This warranty also does not cover physical damage to the product during transit, after purchase of the product but before installation.

The purchaser must request repair or replacement of the defective component through a written notice to the supplier, and must provide the purchaser with the time and opportunity to make such repair or replacement. Otherwise, the supplier will be freed from liability for the defect.

Any transport and exchange costs for the repair or replacement shall be borne by the purchaser. The supplier shall also not be liable for costs incurred in dismantling or fitting replacement parts or for any independent inspection undertaken by the purchaser. The purchaser shall return any allegedly defective goods, postage or freight paid, to the supplier at the address below. Upon receipt of the goods and inspection there of the supplier shall repair or replace, at the supplier's discretion, the defective components and shall return the same to the purchaser, return postage and freight paid. This shall constitute full compliance of the supplier's warranty obligations hereunder.

The seller accepts no liability for the direct or indirect consequences of any modifications of or repairs to the product made out by the purchaser or by a third party without the prior consent of the supplier.

This warranty applies only to the repair or replacement of components of the product and expressly excludes responsibility for damage and consequential damages not occurring to the product itself.

This warranty is the purchaser's exclusive remedy. The supplier expressly acknowledges that it shall not be liable to the purchaser or any client of the purchaser for any direct or indirect damages, injury to persons or property or any consequential damage or loss of profits arising from defective goods or workmanship or from any other cause whatsoever.

To obtain warranty service or to ask any questions regarding the foregoing, please contact our office:

Güntner US LLC
3601 Algonquin Road Suite 925
Rolling Meadows, Illinois 60008
847- 781-0900 Tel
847-781-0901 Fax
dq-us-service@guentner.com or sales.us@Guntner.com
Online support: <https://www.guntner-solutions.us/contact>

2. Important basic information

2.1 Safety instructions

2.1.1 Observing operating instructions

WARNING

Always keep the operating instructions in the unit's immediate vicinity at all times.

Ensure that the operating instructions are accessible to all people that have anything do with the unit, at all times.

Ensure that the operating instructions are read and understood by all people that associated with the unit.

2.2 Responsibilities

2.2.1 Manufacturer's responsibilities

The notes provided in these operating instructions on maintaining the unit's functional safety, preventing possible hazards when transporting, setting up and installing, start-up and operation, and with maintenance activities (cleaning, servicing and repairing) refer exclusively to the unit.

The manufacturer's responsibilities are documented in acc. with local regulations (design, manufacture and testing).

The construction, soldering and welding materials are configured so that they withstand the foreseeable mechanical, thermal and chemical stresses, and are resistant to the heat carrier and the heat carrier/refrigerator oil mixture used.

The heat carrier carrying parts of the unit (core tubes, distributor tube and header outlet) are configured so that they remain tight with the foreseeable mechanical, thermal and chemical stresses, and withstand the maximum permissible operating pressure.

Material, wall thickness, tensile strength, corrosive resistance, shaping process and testing are suitable for the heat carrier used and withstand the possible pressures and stresses that might occur.

All responsibilities regarding the equipment, into which the unit is integrated, are the exclusive responsibility of the people involved in the individual workflows.

2.2.2 Responsibilities of the system's installer

The responsibilities of the system installer, including the heat carrier circuit, are documented in the system's version (design, manufacture and testing – cooling equipment and refrigeration system) in accordance with local regulations.

Component supplier-system installer interfaces:

- Inform Güntner if faults occur:

Inform Güntner immediately if faults occur during the set-up, installation, start-up and operation.

The responsibilities of the system installer in particular include:

- Planning and preparing emergency measures:

To avoid consequential damage caused by operational disruptions, a warning system which immediately signals all faults must be provided on-site. Prepare emergency measures that prevent consequential damage for people and property should faults occur.

- Install emergency STOP switches that can be actuated without danger.
- Specify checking and maintenance intervals:

The system including the heat carrier circuit/ (complete system: must be configured and equipped with all required equipment for maintenance and sufficient servicing and testing in acc. with local regulations.

With the integration of the unit into the refrigeration system's heat carrier circuit, the heat carrier and version must not deviate from the order-related information specified in the order-related offer documents.

The installer of the system including the heat carrier circuit; must refer to the requirement for sufficient instruction of the operating and supervision staff when operating and maintaining the equipment.

It is recommended that the future customer staff – if possible – be present with the set-up and installation, with the tightness test and cleaning, with the filling with heat carrier and with the setting of the equipment including the heat carrier circuit.

2.2.3 Owner or operator responsibilities

The owner or operator responsibilities are documented in the operation, maintenance, servicing and recovery of the system including the heat carrier circuit in acc. with local regulations.

The owner or operator must ensure that the proper people are sufficiently trained and qualified for operating, monitoring and servicing the system including the heat carrier circuit.

The operating personnel for the system and the heat carrier circuit must have sufficient knowledge and experience with regard to the mode of operation, operation and daily monitoring of this system including the heat carrier circuit.

Before the system start-up, including the heat carrier circuit, the owner or operator must ensure that the operating personnel are sufficiently instructed with the system's documentation (which is part of the operating instructions) on the set-up, monitoring, mode of operation and servicing of the system including the heat carrier circuit and the safety measures to be observed, and with regard to the properties and handling of the heat carrier to be used.

The owner or operator must ensure that when operating, monitoring and maintaining the system including the heat carrier circuit the heat carrier and version must not deviate from the details specified in the order-related offer documents.

Planning and preparing emergency measures: To avoid consequential damage caused by operational disruptions, a warning system must be installed on the customer's premises. Prepare emergency measures that prevent consequential damage for people and property should faults occur.

Responsibility remains with the owner or operator of the system, including heat carrier circuit. If the system, including heat carrier circuit, is used by somebody else; there must be an agreement on sharing responsibility.

2.3 Legal notes

Warranty claim expires as follows:

- With faults and damages that can be attributed to non-compliance with the specifications of these operating instructions.
- With complaints that can be attributed to use of spare parts other than the original spare parts specified in the order-related offer documents.
- With changes to the unit (heat carrier, version, function, operating parameters, etc.) vis-à-vis the order-related information specified in the order-related offer documents, without the manufacturer's prior consent.

The operating instructions may not be reproduced electronically or mechanically, circulated, changed passed on to third parties, translated or used otherwise, in full or in part, without Güntner's prior explicit written approval.

2.4 Operating instructions

2.4.1 Scope

These operating instructions apply for all drycoolers of the GFD series.

NOTICE

You will find the precise type of your unit in the attached order-related offer documents.

2.4.2 Set-up and other applicable documents

The unit's operating instructions include the following parts:

- These instructions
- Order-related offer documents
 - The order-related offer documents are attached to these instructions and include the following information:
 - The order-related proper use as specified
 - The order-related scope of delivery
 - The order-related technical data
 - The order-related drawings specifying customer, project number and order number
 - Motor connection wiring diagram in terminal boxes

These operating instructions are part of the operating instructions manual of the system, provided by the system's installer including the heat carrier circuit.

2.5 Conventions

2.5.1 Typographical conventions

The following text markups are used in these operating instructions:

Bold	Requires special attention!
Grey triangle	Instructions

2.5.2 List of abbreviations

Abbreviations	Meaning
EN	European Norm
DIN	German industrial standard (specification of a standard)
ISO	International Standardization Organization
Emergency STOP	Switch for immediately switching off the refrigeration system
°C	Degrees Celsius (Celsius scale temperature)
bar	Unit of pressure
l	Litre (liquid volume)
Vol %	Volume percent (concentration level relative to a volume)
IP	Insulation protection
Q 6.3	Balancing quality
ppm	parts per million, concentration figure, stands for "millionth"
Hz	Hertz (frequency)
3~	3-phase rotating current
1~	1-phase alternating current
VDE	"Verband der Elektrotechnik, Elektronik und
TCC	Informationstechnik" (Association for Electrical, Electronic & Information Technologies)
EPC	Technical Connection Conditions
VDI	Electric Power Company
NEC	"Verein Deutscher Ingenieure" (German engineers' association)
UL	National Electrical Code

2.6 Conventions for safety signs and notices

2.6.1 General safety signs and their meaning in these operating instructions


Dangerous situation that will definitely cause serious injury or death if it is not avoided.

Dangerous situation that could cause serious injury or death if it is not avoided.

Dangerous situation that could cause slight to moderate injury if it is not avoided.

Refers you to possible damage to property.

2.6.2 Warning signs and their meaning in these operating instructions



Warns against hand injuries!

Hands or fingers can be crushed, pulled in or otherwise injured with non-compliance.



Warns against hot surfaces!

The temperature is over +45 °C (protein clotting) and can cause burns.



Warns against dangerous electrical voltage!

Danger of electric shock if voltage-carrying parts are touched.



Warns against fire-risk substances at set-up point.

Use of ignition sources can cause fire at set-up point.



Warns against harmful to health or irritant substances at set-up point

Contact with or inhaling harmful to health or irritant substances can cause injuries or damage the health.

2.6.3 Prohibitory signs and their meaning in these operating instructions



Flames / flammable substances prohibited

Ignition sources must be kept away and ignition sources must not develop!



No smoking!

Smoking is forbidden.

2.6.4 Mandatory signs and their meaning in these operating instructions.



Use eye protection!

Eye protection: Use protective cover, protective glasses or face protection.



Use hand protection!

Protective gloves must protect against mechanical and chemical dangers (see imprinted pictograms).



Use respiratory protection!

Breathing apparatus must be suitable for the heat carrier used. Breathing apparatus must consist of:

- At least two independent breathing devices (self-contained breathing apparatus)



Use protective clothing!

Personal protective clothing must be suitable for the heat carrier used and for low temperatures, and must have good heat insulation properties.

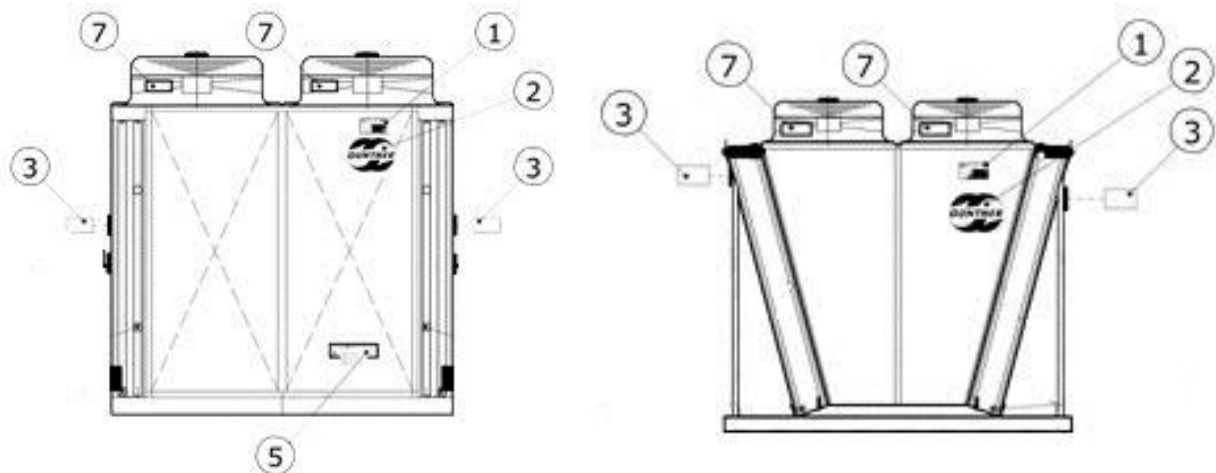


Deactivate before work!

Deactivate the electrical system and secure against switching on again before starting installation, maintenance and repair work.

3 Safety

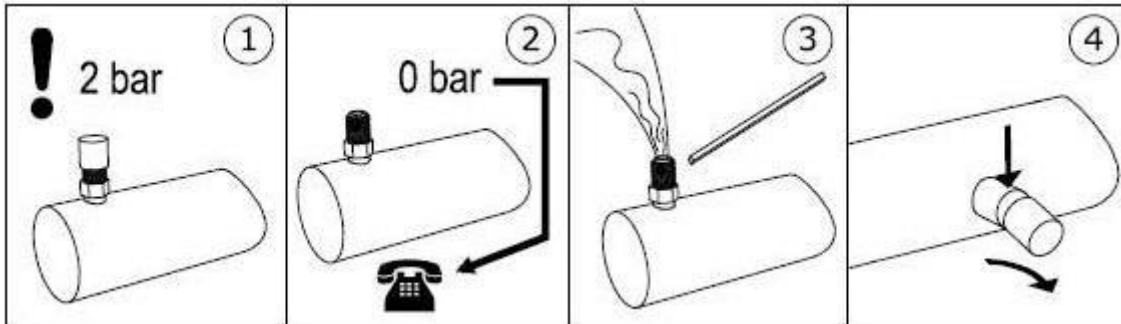
3.1 Labeling on the unit:



1. Unit type name	6. Fan guard" warning sign – only with units with individual removable fan protection, e.g. long throw units, woods fans
2. Logo	7. Fan number label
3. Transportation filling warning sign next to Schrader valve	8. Transportation regulation on the outside
4. Transportation regulation	9. Connections ON, OFF
5. Inspection door, only on units with inspection doors	10. Caution! Operating pressure 17 bar

3.1.1 Safety signs on the unit

Safety signs on the unit individually



"Transportation filling" warning sign next to Schrader valve

Achtung • Caution • Attention

Ventilator-Betrieb nur mit Schutzgitter

GEFAHR!

Fan operation only with guard grille!

DANGER!

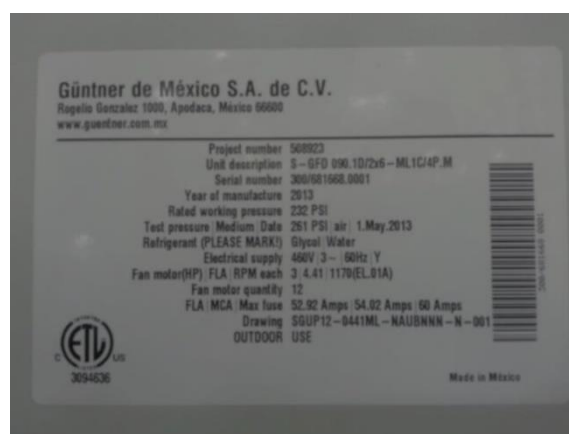
Mode operation de ventilateur qu'avec grille protective!

DANGER!

"Guard grille" warning sign (only used with separate removable guard grille)

3.1.2 Other signs and notes on the unit

1. Unit type plate



2. Güntner logo



3. Connections

Outlet / Salida**Inlet / Entrada**

3.2 Basic safety notices

3.2.1 How to act in an emergency

 WARNING**Danger of injuries and damage to property!**

The unit is operated as standard with the heat carrier, ethylene glycol (synonyms: ethane-1,2-diol, glycol). When using other heat carriers, prior agreement with the manufacturer is an absolute must.



Ethylene glycol is a colorless, slightly viscous, not very fluid, water-mixable, deliquescent liquid with a sweet smell or taste. Ethylene glycol is flammable and explosive at higher temperatures in a vapor/gaseous state.

With skin contact ethylene glycol causes slight irritation with the danger of skin absorption; with eye contact mucous membrane irritation; with ingestion agitation with disruption of the central nervous system; with prolonged endangering effect – fatigue, coordination disruption, unconsciousness, kidney damage.

⚠ WARNING



Keep ethylene glycol away from ignition sources. No smoking!

Ethylene glycol vapors are heavier than air and may flow off to rooms on a lower level. In still air there may be an increase of the ground level concentration. With high concentrations, there is a danger of suffocation due to reduced oxygen concentration, especially at ground level.



Avoid contact with skin, floor, and clothing! Remove contaminated, soaked clothing immediately!

Do allow contact with heavy oxidation agents (chromo-sulphuric acid, potassium permanganate, smoking sulphuric acid, or similar)! Danger of strong reactions! Unauthorized people must not have access to the unit. When setting up it must be ensured that the heat carrier escaping from the unit cannot penetrate the interior of the building or put people at risk in any other way.



Safety measures and procedure:

- With unexpected serious heat carrier breaks, activate the emergency STOP switch set up in a safe place, e.g. with:
 - Visibly escaping heat carrier from the heat exchanger or pipe components.
 - Sudden large release (release and evaporation of the greater part of the entire heat carrier/ in a short time, e.g. in less than 5 minutes).
- Have experienced, trained personnel with prescribed protective clothing perform all necessary protective and other measures:
 - Use respiratory protection.
 - Use a room air-independent breathing apparatus with maintenance work in high heat carrier concentrations in the room air.
 - Ensure the set-up room is well ventilated.
 - Divert escaped heat carrier liquid safely in acc. with current regulations.

3.2.2 Personnel, care requirements

WARNING

The unit must only be put into operation, operated, maintained and repaired by trained, experienced and qualified personnel. People that are responsible for the operation, maintenance, repair and evaluation of systems and their components must have the required training and specialist knowledge for their work in acc. with current regulations to be qualified. Qualified or expert means the ability to satisfactorily perform the activities required for the operation, maintenance, repair and evaluation of refrigeration systems and their components.

The unit may be operated by operating personnel that have no specific knowledge of refrigeration engineering, but have sufficient knowledge and experience with regard to the mode of operation, operation and daily monitoring of this system, including the heat carrier circuit. These operating personnel may not make any interventions or settings on the system including the heat carrier circuit.

Changes to the unit, which the manufacturer has first agreed to in writing, may only be made by the instructed and qualified personnel.

Electrical installation:

Work on the electrical equipment may only be performed by personnel that have the required expertise (e.g. an electrician or an electro-technically instructed person), and who are authorized by the operator, in compliance with the respective VDE regulations (and national and international provisions) and the TCCs of the EPCs.

3.3 Proper intended use

3.3.1 Proper intended use

Drycoolers of the GFD series are intended for installation in the heat carrier circuit of a refrigeration system and for outdoor installations. The drycoolers cool the heat carrier by giving off heat to the air that is being moved by fans over the dry heat exchanger surface.

The unit is delivered for operation with a specific operating point:

- Heat carrier flow temperature
- Heat carrier return temperature or circulating heat carrier volume
- Airflow volume
- Air inlet temperature

The specified operating point is provided in the order-related offer documents.

3.3.2 Operating conditions

The unit is a component of the heat carrier circuit a cooling system including its heat carrier circuit. The purpose of these operating instructions, as part of the operating instructions manual (of which these operating instructions are a part), is to restrict the dangers to people and property and the environment from the unit and the heat carrier used in it to a minimum. These dangers are essentially connected with the physical and chemical properties of the heat carrier and with the pressures and temperatures that occur in the heat carrier-carrying components of the unit (see Residual hazards with the heat carrier section, page 20).

⚠ WARNING

Danger of injuries and damage to property!

The unit must only be used in acc. with the proper intended use. The operator must ensure that when operating, monitoring and maintaining the unit, the fluid and version do not deviate from the order-related information specified in the order-specific offer documents.

The operator must ensure that maintenance measures are performed in compliance with the system's operating instructions manual.

Filling the unit is only permitted following written approval by the manufacturer. You will find the order-related proper use as intended in the order-specific offer documents.

Do not exceed the max. Operating pressure given on the unit's type plate.

3.3.3 Improper use

⚠ WARNING

Danger of injuries and damage to property!

Heat carriers and their combinations with water or other substances in the heat carrier-carrying components have chemical and physical effects from the inside on the materials surrounding them. The unit must only be pressurized with heat carrier glycol. Pressurizing the unit with another heat carrier results in,

Material, wall thickness, tensile strength, corrosive resistance, shaping process and testing are not suitable for the heat carrier used and do not withstand the possible pressures and stresses that might occur.

The unit not being resistant to the other heat carrier and the other heat carrier mixture.

The unit not remaining tight during operation and when shut down.

A possible sudden escaping of heat carriers could directly endanger people and/or property and the environment.

The maximum permissible operating pressure specified on the type plate must not be exceeded! If the operating pressure is exceeded,

The structural and welding materials will not withstand the foreseeable mechanical, thermal and chemical stresses and the pressure that can occur during operation and when shut down.

The unit will not remain tight during operation and when shut down.

There may be a possible sudden escaping of heat carriers after a break or leakage on heat carrier-carrying components, which would result in the following dangers:

- Danger of escaping materials
- Suffocation hazard
- Hazards caused by panic reactions
- Environmental pollution

⚠ WARNING

Drycooler may not be used

- Where it is possible that short or prolonged effect caused by contact, inhalation or ingestion of the heat carrier glycol might result in harmful hazards.
- Where the possibility exists of a sudden large release of the greater part of the entire heat carrier filling in a short time (e.g. in less than 5 minutes).
- The unit must not be changed without prior written consent by Güntner. Changes to the unit are:
 - Changing the operating point (see chapter Unit)
 - Changing the fan capacity (air volume)
 - Changing the heat carrier flow-through volume
 - Changing over to another heat carrier

The unit must not be operated if safety devices recommended by the manufacturer are not available, not properly installed or not fully functional.

The unit must not be operated if it is damaged or demonstrates faults. All damages and faults must be reported to Güntner immediately and must be removed immediately.

Work on the unit must not be performed without the personal protective equipment specified in these operating instructions.

3.4 Mechanical residual hazards

2.4.1 Fins, sharp unit corners and edges

⚠ WARNING



Warns against hand injuries!
Danger of cuts on hands and fingers from the fins and sharp corners and edges of the unit.



Use reliable hand protection!

3.4.2 Fans

⚠ WARNING	
	Danger of cutting off, pulling in! There is a danger of cutting off fingers on the rotating fan blades, injury hazard for the hands and pulling in danger for loose elements such as hair, necklaces or clothing parts.
	Do not operate fans without guard grille. Pinch/trap point hazard! With automatic fan start during maintenance work there is a danger of pinching/trapping for the hands and fingers.
	Power off the unit before you begin maintenance work with which you must remove the guard grille.

3.5 Electrical residual hazards

⚠ WARNING	
	Warns against dangerous electrical voltage! Direct and indirect contact with voltage-carrying parts of motors and electrical lines can cause serious injuries or death.
	Power off the unit before you begin maintenance work. See the refrigeration system's system documentation for this. Please note that the mains cables may also be carrying voltage, even if the unit is powered off. Work on electrical equipment must only be performed by people that have the required expertise (e.g. an electrician or an electro-technically instructed person) and who are authorized to do so by the operator.

3.6 Thermal residual hazards

3.6.1 Danger of burns

 WARNING
Warns against hot surfaces! In operation the unit's heat exchanger and pipes have temperatures of over +45 °C. Contact can cause burns. With heat carrier temperatures over +45 °C there is the hazard of burns on pipe components and distribution pipes. Use hand protection!

3.6.2 Freeze hazard

 WARNING
Warns against freezing! With insufficient frost protection filling and operation in areas with frost hazard the unit can cause freezing in the unit. With units that cannot be drained completely the hazard of freezing also remains after draining. It is imperative that the proper ventilation be ensured when draining the unit. With pressure test, operation and shutdown of units filled with water or insufficient frost protection filling or insufficient setting of the heat carrier, these are disrupted at minus temperatures.

3.7 Residual hazards with the heat carrier

 WARNING
Danger of injuries and damage to property! General description of hazards: The unit is operated as standard with the heat carrier ethylene glycol as the heat carrier (synonyms: ethane-1,2-diol, glycol) . When using other heat carriers prior agreement with the manufacturer is an absolute must. Ethylene glycol is a colorless, slightly viscous, not very fluid, water-mixable, deliquescent liquid with a sweet smell or taste. Unauthorized people must not have access to the unit. When setting up it must be ensured that the heat carrier/ escaping from the unit cannot penetrate the interior of the building or put people at risk.

⚠ WARNING

Danger of harm to health!

- With skin contact ethylene glycol causes slight irritation with the danger of skin absorption; with eye contact mucous membrane irritation; with ingestion agitation with disruption of the central nervous system; with prolonged endangering effect – fatigue, coordination disruption, unconsciousness, kidney damage.
- Avoid contact with skin, eyes and clothing! Remove contaminated, soaked clothing immediately!
- Ethylene glycol vapors are heavier than air and may flow off to rooms on a lower level. In still air there may be an increase of the ground level concentration. With high concentrations, there is a danger of suffocation due to reduced oxygen concentration, especially at ground level.
- To avoid the inhalation of high vapor concentrations, the working rooms must be well aired.
- Test the tightness of the unit regularly, as specified in these operating instructions.



Ignition and fire hazard!

Ethylene glycol is flammable and explosive at higher temperatures in a vapor/ gaseous state.

Keep ethylene glycol away from ignition sources.

Suitable fire fighting equipment must be provided on site with work involving fire or sparks, such as grinding, welding, etc.



Ensure that the provided fire fighting equipment is provided in sufficient quantities, that it functions properly and that the extinguishing agent does not react with the heat carrier.

No smoking!



Danger of poisoning!

Contact of the heat carrier with open fire must be prevented, as toxic combustion products can form.

Prevent heat carrier contact with open fire!

Welding and soldering must therefore only take place after completely draining the relevant section of the system of the heat carrier. Ensure good ventilation here!



Do not bring into contact with heavy oxidation agents (chromo-sulphuric acid, potassium permanganate, smoking sulphuric acid, or similar)! Danger of strong reactions!

Frostbite hazard!

With insufficient frost protection filling and operation in areas with frost hazard the unit can cause freezing. With units that cannot be drained completely the hazard of freezing also remains after draining.



It is imperative that the proper ventilation be ensured when draining the unit.

With pressure test, operation and shutdown of units filled with water or insufficient frost protection filling or insufficient setting of the heat carrier, these are disrupted at minus temperatures.

3.8 Residual hazards caused by vibrations

WARNING

Danger of injuries and damage to property caused by escaping materials

If fans are damaged during fan operation, flying parts of the fan blades can injure people or cause damage to property close to the fan.

Fans, components and cables in the system, including the heat carrier circuit must be designed, constructed and integrated so that dangers caused by vibrations that it or other parts of the system generate are reduced to an absolute minimum, while incorporating all available means for reducing vibrations, preferably at the source.

NOTICE

Damage to property caused by vibrations

Vibrations that are increased by imbalances, as created by dirt or fan blade damage, are regularly caused with fan operation. The vibrations are transferred to the unit, where they can cause damage and damage the unit mounting or heat carrier circuit components connected to the unit.

Check the fan blades and guard grille regularly for dirt and frosting and/or icing and the fans' smooth running (see Fans section, page 47).

Check the fan blades and guard grille regularly for dirt and the fans' smooth running (see Fans section, page 47).

3.9 hazards caused by pressurized parts

WARNING

Injury and damage to property caused by pressurized components that contain a heat carrier!

Breaks in pressurized pipes or pressurized components of the unit can cause injuries or damage to property caused by escaping materials. A sudden large release of the heat carrier with its hazardous properties after a break or leak on pressurized components of the unit can cause the following hazards:

- Irritations and health hazards
- Strong chemical reactivity
- Flammability
- Suffocation
- Panic,
- Environmental pollution

Ensure that the unit in question is pressure-free before maintenance work begins or remove the heat carrier from the unit in question.

Only perform maintenance work – especially soldering work – on the unit in question after completely removing the heat carrier from the unit.

3.10 Residual hazards caused by defective installation

WARNING

Injuries and damage to property caused by defective installation! Defective installation results in hazards caused by:

- Break or leak on liquid-carrying unit components and pipes
- Uneven load distribution on the fixtures with the danger of stresses within the unit or unit displacement (breaks or leaks on fluid-carrying components of the unit and pipes; danger of breaking off).
- Insufficient securing of heat carrier-carrying lines against mechanical damage! On-site connections: loaded installation; effect of forces on the distribution and header pipes with the danger of breaks or leaks on fluid-carrying components of the unit and pipes; danger of breaking off!
- Break-off and fall danger of the unit with hazard of escaping heat carrier and exposed electrical cables.
- Break-off danger of the unit with hazard of escaping heat carrier and exposed electrical cables.
- Danger of damage caused by environment-conditional hazard sources (production, transport and other processes at the set-up point).
- Unit functional faults caused by air inlet/outlet obstructions.
- Obstruction of all-side inspection, checks and maintenance, i.e. no unobstructed accessibility to the heat carrier-carrying and electrical components, connections and cables, no recognizable identifiers on the pipes and insufficient space for tests.

Ensure that:

The units are to be installed on the fixing points corresponding with their weights and tightened with fixing bolts. The operator or installer is responsible for ensuring that the bolted connections are of an adequate strength.

The diameters of the mounting holes have been statically determined by the manufacturer and the fixing bolts are adapted accordingly.

The fixing bolts are secured against loosening by means of an appropriate locking device.

- The fixing bolts are not over-tightened or stripped.
- All fixing bolts are tightened equally to achieve a load distribution on the connections that is as balanced as possible.
- All fixing points maintain the same spacing to the fixing level permanently and under load, so that no mechanical stress occurs in the unit structure. The units are anchored in their fixing position in order to prevent the equipment from moving.
- The functional safety of the fixing bolts is tested as part of the maintenance periods. See Maintenance section, page 45.
- The unit is fixed and set up so that it is not damaged by environment-conditional hazard sources (production, transport and other processes at the set-up point) or its functioning is not disturbed by the interventions of unauthorized persons.
- The units are fixed and set up with sufficient slopes for drip water flow.
- The units are fixed and set up so that unobstructed air inlet/outlet is constantly available without any air short circuiting.
- The units are fixed and set up so that unobstructed heater rod swap-out with electric defrosting is constantly available (option: accessory at customer's request).

⚠ WARNING

The units are fixed so that they can be inspected, checked and maintained from all sides at all times, i.e. there must be unobstructed access to the refrigerant-carrying and electrical components, connections and lines, the pipeline labeling must be identifiable and adequate space must be available for testing.

The heat carrier-carrying lines are protected against mechanical damage. On-site connections: when installing keep the unit free of load; force must not be exerted on the distribution and header pipes.

The following must be observed without fail when installing the unit:

- Imperative adherence to spacing from objects that could be endangered.
- Easily flammable materials must not be placed below the unit.
- Set up and fix units as follows: In areas that are used for inner-plant traffic, the pipelines to and from the unit must only be installed with connections and fittings that cannot be removed.
- Release devices to prevent liquid escapes must be provided and available.

3.11 Residual hazards with break during operation

⚠ WARNING

Injuries and damage to property caused by break during operation!

- Residual hazards with break during operation (see Residual hazards caused by defective installation section, page 22),
- Non-compliance with maximum permissible operating pressure (see Operating conditions section, page 15),
- Disregarding pressurized line sections with maintenance (see Residual hazards caused by pressurized parts section, page 21).
- Disregarding residual hazards caused by vibration (see Residual hazards caused by vibrations section, page 21) result in ruptures during operation and maintenance. This results in dangers caused by:

- Escaping materials (see Residual hazards caused by pressurized parts section, page 21).
- Released heat carrier (see Residual hazards with the heat carrier section, page 20). Ensure that:

- The installation is fault-free.
- The maximum permissible operating pressure is always adhered to.
- Pressurized line sections are depressurized before all maintenance and repair work.
- Vibrations from the refrigeration system (vibrations caused by refrigeration system compressors, components and lines) and from the fan (imbalances caused by frosting, icing or dirt build-up or damages) are reduced with all available means and brought down to an absolute minimum.

- Vibrations from the refrigeration system including the heat carrier circuit, (vibrations caused by the complete system's compressors, components and lines, including the heat carrier circuit) and from the fan (imbalances caused by frosting, icing or dirt build-up or damages) are reduced with all available means and brought down to an absolute minimum.

- Release devices to prevent liquid escapes are provided and available.

3.12 Residual hazards caused by escaping objects or liquids

 WARNING
Injuries and damage to property caused by escaping objects or liquids! Residual hazards caused by escaping objects and liquids (see Residual hazards with break during operation section, page 23).

3.13 Residual hazards with disposal

 WARNING
Danger of injuries and damage to property caused by heat carrier glycol! The following notes are recommendations for the proper professional disposal of the unit. Applicable waste disposal laws are binding for the country of operation: ■ Disposal must only be carried out by experts. ■ All unit components, e.g. heat carriers, heat exchangers, fans, etc. must be disposed of properly as specified. ■ Used heat carrier that is not determined for reuse, must be treated as waste and safely disposed of. There must be no emissions into the environment. ■ A disposable single-use container must not be used, as it is possible that heat carrier residues in the container may escape during disposal. ■ The heat carrier container must not be overfilled. The maximum permissible pressure of the heat carrier container must not be exceeded during the work process. ■ The heat carrier must not be filled in a liquid container that contains another or an unknown heat carrier. This other or unknown heat carrier must not be released into the atmosphere, but rather identified, treated again, or properly disposed of as specified. ■ An officially authorized facility can be used for destroying the heat carrier. ■ It must be ensured that all unit components containing heat carriers are disposed of properly as specified. ■ The unit consists predominantly of the basic materials, copper, aluminum, galvanized steel (heat exchangers and enclosures), steel, polyamide (motors). These materials can be handled by the waste industry, including in paint-treated state, to recycling via mechanical and thermal separation. ■ Before scrapping the heat carrier-carrying unit components must be drained, whereby the pressure must be reduced to at least 0.6 bar absolute for a unit pipe volume up to and including 200 l and to 0.3 bar absolute for a unit pipe volume over 200 l. The pressure reduction process is then ended when the pressure no longer increases and remains constant, and the unit is at ambient temperature.

 WARNING
Danger of environmental pollution! Ensure that no heat carrier enters water systems or sewage. Operate the facility for recovery or disposing of the refrigerant, so that the danger of a heat carrier emission into the environment is kept as low as possible. Güntner's transportation packaging is made from environmentally compatible material and is suitable for recycling.

4 Technical data

4.1 Unit

NOTICE

The fans' capacity values depend on the ambient temperature and on the air resistance at the set- up point.

Güntner recommends electric fan ring heating with use in the temperature range.

Please consult the manufacturer when operating the unit below -40°C because of the special material requirements and selection.

All electrical parts must be installed in acc. with NEC and UL standards.

Project number	See order-related offer documents
Unit name	See order-related offer documents
Manufacturer number	See order-related offer documents
Production year	See order-related offer documents
Heat carrier	See order-related offer documents
Volume	See order-related offer documents
Permissible operating pressure	16 bar
Test pressure	17.6 bar
Permissible air humidity	100 %
Test date	See order-related offer documents
Test medium	Dry air
Airborne noise emitted	See order-related offer documents In acc. with standard procedure for calculating sound level in acc. with EN 13487; Annex C (normative).
Weight	See order-related offer documents

4.2 Fans

Güntner control units can be used for speed-controlling GFD fans used. Rotating current fans can generally be operated with step control or variable speed control. 3 sound levels (N, M, L) can be delivered.

Fan type	See order-related offer documents
Protection rating	IP 54, thermal class 155 (F) and DIN VDE 0530
Current type	Rotating or alternating current
Voltage	230 V 3 ~ 60 Hz / 460 V 3 ~ 60 Hz
Balancing quality	Q 6.3 in acc. with VDI 2060
Permissible air temperature	Usage range: -30 °C to +55 °C
Fan type	See order-related offer documents
Protective devices	• Thermal: Thermo contacts (break contact protective device for preventing thermal overload). • Mechanical: Protective contact grille in acc. with EN 294

5 Set-up and function

The drycooler consists of a heat exchanger, consisting of pipe coils fitted with fins (copper pipe, aluminum fins), distribution and header pipes (copper) and pipe connections to the pipeline system

And – depending on the version – with one or more low noise axial fans with maintenance-free motors.

The refrigeration system is a combination of refrigerant-carrying components and fittings connected with one another, which form a closed circuit, in which the refrigerant circulates.

The GFD series drycoolers feature reduced installation input with:

- Fewer unit legs with Güntner support profiles, therefore fewer bases are required.
- Lower roof load with reduced unit weight.

The heat carrier is the heat carrier that is used for the heat exchange, and which remains in the liquid phase during the heat absorption.

The liquid used is a heat carrier in acc. with current regulations. There is no imminent danger for the staff.

The heat carrier is used for the heat exchange, and which remains in the liquid phase during the heat absorption. The drycooler cools the heat carrier by giving off heat to the air (air- side) that is being moved by fans over the dry heat exchanger surface.

Fan motor

The fan motors can be speed-controlled or step-controlled with Güntner control units. The direction must be checked; a change in the direction, if the direction is wrong, is made by interchanging two phases.

6 Fan motor

NOTICE

During longer storage or downtime periods, the fans must be operated for 2 to 4 hours each month.

NOTICE

In the case of fans with protection type IP55 or higher, any sealed condensate drain holes must be opened at least every six months.

EC technology

No maintenance required.

AC technology

The AC motors are protected against overheating by a thermo-contact (or positor). For motors with thermocontacts, the thermocontact has to be wired in the switch cabinet in such a way that turning on of the motor with triggered thermocontact is not possible. A locking device is recommended for preventing reactivation.

Motors with PTC resistors require an additional trigger device for the installed thermistors. Locking is recommended to prevent reactivation. Max. 2.5 V test voltage or current-limited meters may be used on thermistors.

With star-delta connection, corresponding time delays must be taken into consideration.

For motors with direct start and a connection value > 4.0 kW, a startup current limitation (soft start using thyristor) may be necessary.

If frequency converters for speed control are used, the following has to be observed for external rotor motors:

Between frequency converter and motor, an effective all-pole sine filter has to be installed (output voltage filter effect between phase against phase, phase against earth).

Güntner frequency converters feature this function as standard. Güntner three-phase standard motors are suitable for direct operation on frequency converters.

The three-phase fan motors can be operated by means of star-delta connection with two speeds and/or with speed control. The direction of rotation must be checked. If the direction is wrong, it can be changed by interchanging two phases.

Nema Motors

NOTICE

For maintenance of Nema motors, please consult the supplier's recommendations.

7 Transportation and storage

7.1 Safety

⚠ WARNING

Crushing danger with falling down!

The unit weighs between 830 kg and 3440 kg. It can slip and fall off the means of transport, causing serious injuries or death. Heavy impacts or vibrations can damage the unit.

Observe the instructions on the transport labels on the packed units. Ensure that the assigned staff is trained for proper unloading.

Use a transporting device appropriate for the unit's weight (see Transportation and storage section, page 27). You will find the weight of the packed unit in the order-related offer documents. Ensure that nobody is under the unit or near the loaded area during the transport.

Observe even distribution of unit weight for transport. Observe the instructions on the transport labels on the packed units (see Other signs and notes on the unit section, page 12).

⚠ WARNING

Secure the unit against slipping and mechanical damage.

When transporting by crane: The hooks and lifting gear of the load lifting equipment must be only attached at the points specified by the manufacturer. Ensure that the unit enclosure is not crushed by slings.

Use auxiliary transport equipment where required. Use a transporting device appropriate for the unit's weight. You will find the weight of the unit in the order-related offer documents (see Set-up and other applicable documents section, page 8). Do not use connection pieces and header pipes as hooking points for lifting, pulling, fixing or mounting. This can cause leaks.

Transport the unit carefully. Avoid setting the unit down hard in particular.

Only step on to the units with firm, tough and safe footwear, and only when required.

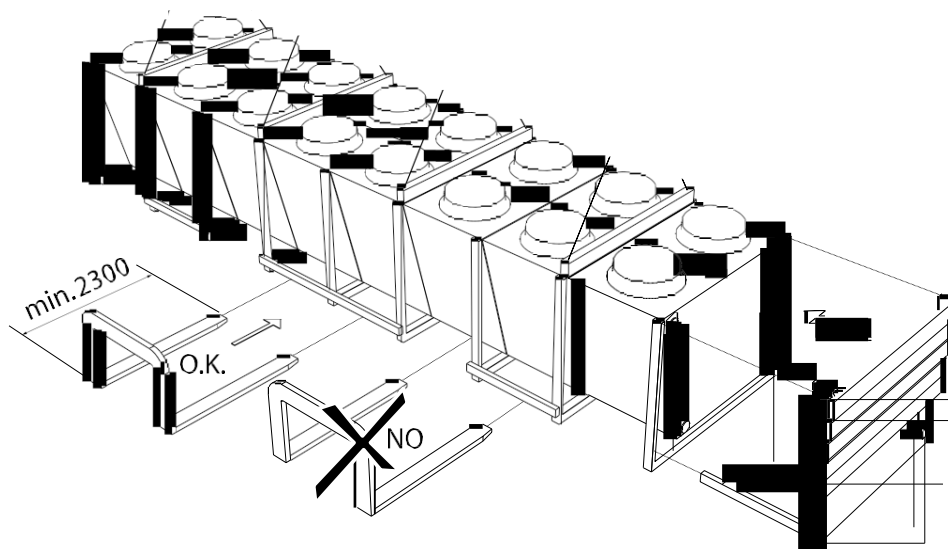
If the units do not have any railings, always only walk on them with a safety device that prevents falling.

7.2 Transportation and storage

NOTICE

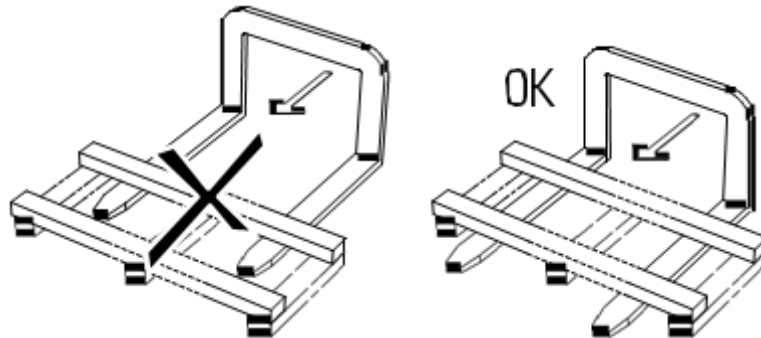
Prolonged mechanical stresses caused by uneven road surfaces and potholes and vibrations during transport by ship can cause transportation damage. Before transportation by sea or in countries with difficult transport routes, attachment parts that are likely to vibrate

- in particular fans and base stands
- must be removed for transportation.
- Transporting the unit at the set-up point
- Unloading the unit



Transport and unload the packed unit with suitable transport equipment (e.g. forklift, crane) at the set-up point.

CAUTION: When transporting with a forklift: Only lift the packed unit with a forklift with full fork length.



7.3 Storage before installation

NOTICE

Danger of corrosion and dirt build-up!
Moisture and dirt must be prevented from entering the unit.

Protect the unit against dust, dirt, moisture and wetness, damage and other harmful effects.

Harmful effects (see Set-up and other applicable documents section, page 8).
Do not store the unit for longer than necessary. Only store the units in their original packaging until installation. Always only place packaging units of the same size on top of one another.

Store the unit at a protected place free of dust, dirt, moisture and damage-free until its set-up (well-ventilated halls or roofed storage site).

If the unit set-up is delayed with regard to the planned installation time: protect the unit against weather and other harmful effects and dirt and other contaminants with an appropriate cover. The unit must also be well-ventilated here.

8 Set-up and start-up

8.1 Safety

8.1.1 Safety instructions for set-up and start-up

⚠ WARNING

Danger of injuries and damage to property with escaping heat carrier!
Incorrect installation causes the danger of heat carrier escaping when the unit is operated and injuries or damage to property (see Residual hazards with the heat carrier section, page 20).

Follow the set-up instructions in this chapter precisely and apply extreme care!

NOTICE

Damage to the system's cooling equipment!

Foreign materials and contaminants in the heat carrier circuit can impair the effectiveness or damage components. Particularly harmful contaminants are:

- Moisture
- Atmospheric air
- Soldering residues
- Rust
- Soot/ash/cinders
- Metal cuttings
- Dust and dirt of all kinds

Moisture in the heat carrier-carrying components of the unit can have the following consequences:

- Ageing and heat carrier decay

Atmospheric air and other non-condensable gases can have the following consequences:

- Heat carrier/ oxidation
- Heat carrier/ chemical reactions

Heat carrier/ chemical reactions with the absence of moisture or atmospheric air with ageing and decay can have the following consequences:

- Formation of organic and inorganic acids
- Corrosion

Other contaminants can cause:

- Accelerated chemical processes (decomposition)
- Mechanical and electrical faults in the heat carrier circuit/

Ensure with installation (connecting the heat carrier-carrying components of the unit to the heat carrier-carrying system of the installation's – or to the refrigeration system's heat carrier circuit) that internal contamination is strictly avoided.

Perform the installation with extreme cleanliness.

Finish all on-site pipe installation work before releasing the transport pressure!

Only release the transport pressure on the Schrader valve immediately before installation.

Only remove the sealing caps on the distribution and header pipe immediately before installation.

NOTICE

Danger of corrosion and dirt build-up!

Humidity and dirt may not get into the unit's interior. If humidity or dirt get into the unit's interior, fittings and other components of the refrigeration installation can be damaged.

Protect the unit against dust, contamination, moisture and wetness, damage and other harmful influences. Harmful influences are, for example:

- Mechanical: Damages caused by impacts, objects falling on or against, collisions with transport equipment, etc.
- Physical: Damages caused by close by concentrated flammable gases
- Chemical: Damages caused by contaminated atmospheres (salt, acid, chlorine, sulphur-containing, or similar)
- Thermal: Damages caused by close by heat sources Start as soon as possible with installation.

⚠ WARNING

The electrical installation must only be performed by electricians in compliance with the relevant VDE rules (or applicable national and international regulations) and the TCCs of the EPCs!

8.1.2 System-side safety requirements

The unit is a component of the heat carrier circuit of an installation and can only be operated in conjunction with the installation.

- All equipment required for operating the unit must be integrated into the switching and activation equipment:
 - Electrics: Fans
 - Heat carriers: valves and fittings
 - An emergency STOP switch that can be actuated without danger must be installed.
 - The heat carrier-side and electrical connections for the unit must be available for the unit on the heat carrier circuit of the refrigeration system. The connections must be specified in the order-related offer documents.
 - The power supply of the fans must be provided in acc. with the specifications on the name plate on the fan motors.
 - A switch-off device for preventing unexpected start-up (repairs switch), which separates all active conductors from the power supply (all-pole switch-off), must be provided for the fans in acc. with NEC.
 - The fans' switch-on/off device must be secured (e.g. with a padlock) to prevent uncontrolled fan start-up.
 - The electrical motor, repairs switch, terminal box and switching cabinet connections must be provided in acc. with the respective connection diagrams.
 - It must be possible to shut off the unit if a leak occurs.
 - People wearing ambient air-independent breathing apparatus in full protective clothing must also be able to activate all safety-relevant shut-off fittings.
 - It must be possible to activate all devices meant for diverting escaping heat carriers from a safe position.

8.1.3 Customer-side safety precautions

⚠ WARNING	
	Danger of injuries! The unit contains a heat carrier (see Residual hazards with the heat carrier section, page 16). The unit is operated as standard with the heat carrier, ethylene glycol as the heat carrier (synonyms: ethane-1,2-diol, glycol). Prior agreement with the manufacturer is imperative when using other heat carriers.
	Ethylene glycol is a colorless, slightly viscous, not very fluid, water-mixable, deliquescent liquid with a sweet smell or taste. With skin contact ethylene glycol causes slight irritation with the danger of skin absorption; with eye contact mucous membrane irritation; with ingestion agitation with disruption of the central nervous system; with prolonged endangering effect – fatigue, coordination disruption, unconsciousness, kidney damage.
	Ignition and fire hazard! Ethylene glycol is flammable and explosive at higher temperatures in a vapor/gaseous state. Keep ethylene glycol away from ignition sources. No smoking! Ethylene glycol vapors are heavier than air and may flow off to rooms on a lower level. In still air there may be an increase of the ground level concentration. With high concentrations, there is a danger of suffocation due to reduced oxygen concentration, especially at ground level. Avoid contact with skin, eyes and clothing! Remove contaminated, soaked clothing immediately! There is no imminent danger for the staff. Ethylene glycol vapors are heavier than air and may flow off to rooms on a lower level. In still air there may be an increase of the ground level concentration. With high concentrations, there is a danger of suffocation due to reduced oxygen concentration, especially at ground level. Do not bring into contact with heavy oxidation agents (chromo-sulphuric acid, potassium permanganate, smoking sulphuric acid, or similar)! Danger of strong reactions! Unauthorized people must not have access to the unit. When setting the unit up it must be ensured that the coolant escaping from the unit cannot penetrate the interior of the building or put people at risk in any other way.

⚠ WARNING

Danger of environmental pollution!

- Set up the unit so that heat carrier, which can escape from the unit in the event of a fault, cannot enter water systems or sewage.
- Operate the facility for recovering or disposing of the heat carrier so that the danger of a heat carrier emission into the environment is kept as low as possible.

8.2 Requirements at the set-up point

You will find the dimensions and weights in the order-related offer documents.

Ensure that the unit is set up so that it is not damaged by environment-conditional hazard sources (production, transport and other processes at the set-up point) and its functioning is not disturbed by the interventions of unauthorized persons.

Position the unit so that it cannot be damaged by internal traffic or transport processes. Enable optimum unit control and accessibility:

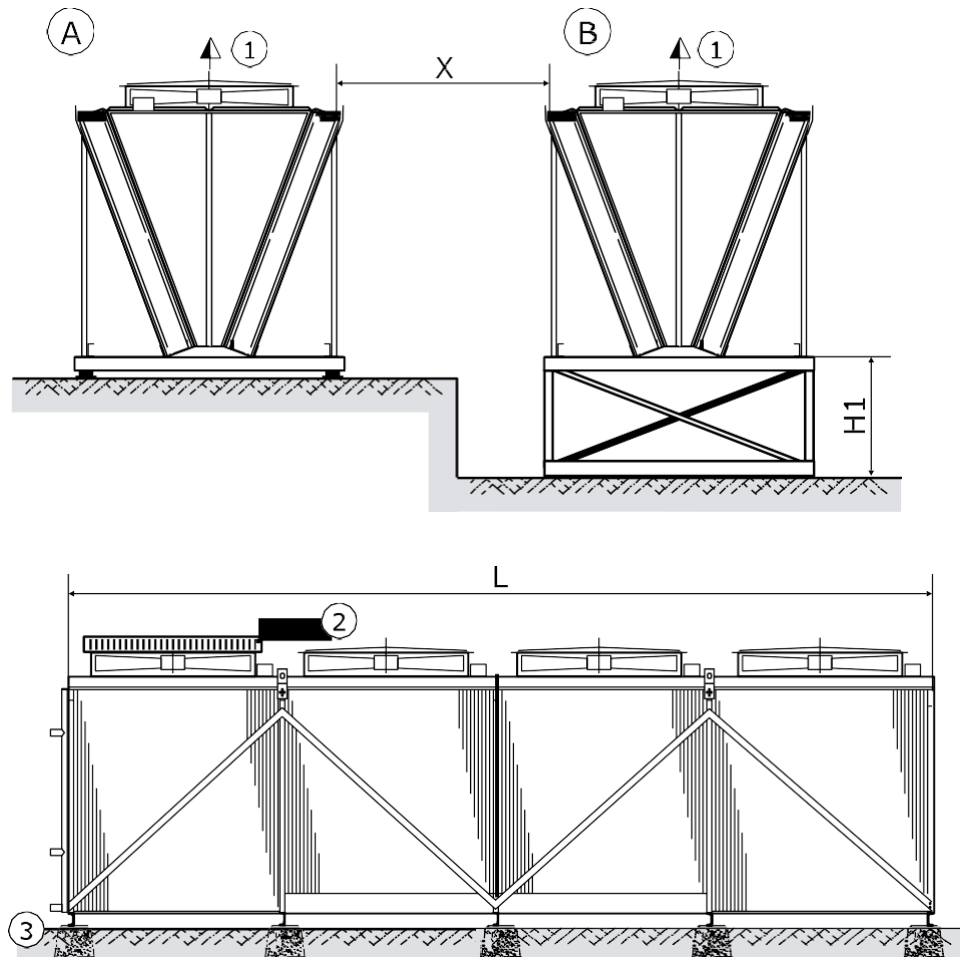
- Place the unit so that it can be monitored and controlled from all sides at all times.
- Ensure that sufficient space is provided for maintenance.
- Ensure that all liquid-carrying components, connections and lines and all electrical connections and lines are easy to access.
- Ensure that the pipes' identification is well visible.

NOTICE

Lightning rod beside the unit are recommended as lightning protection.

Several units can be placed space-saving in rows with high nominal capacity to cover high capacities. With double or multiple row set-ups a substructure is required to guarantee sufficient air-side pressurizing.

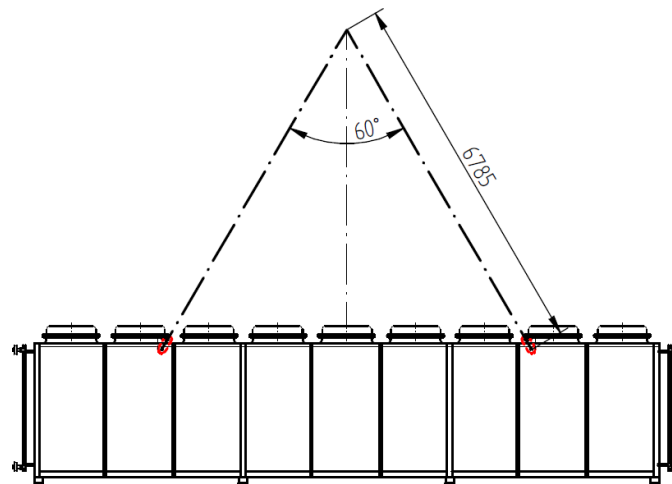
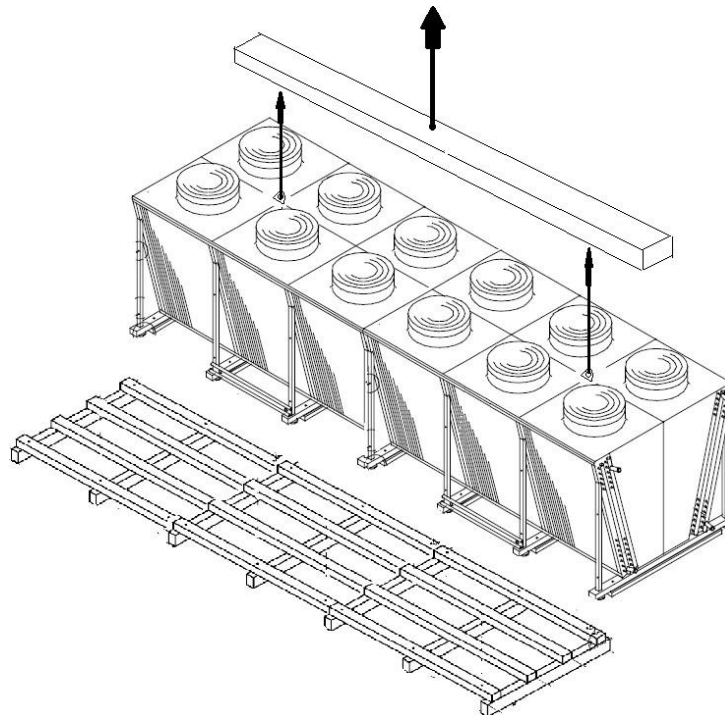
Requirements at the set-up point



- 1 Airflow direction
- 2 Güntner Streamer
- 3 With two or more long side set-up units

Variant A: Set-up at floor level	Variant B: Set-up on a substructure
Min. spacing $X \geq 0.5 * L$ $X \geq 0.35 * L$ (with Güntner Streamer) Spacing X can be reduced by offsetting the units in the lengthways direction. Appropriate definitions must be made in agreement with the manufacturer.	Min. spacing X is reduced depending on height H1. Appropriate definitions must be made in agreement with the manufacturer.

8.3 Rigging and unpacking instructions at the set-up point



8.3.1 Rigging instructions

An appropriate crane is required to unload the unit from the flat bed.

1. Insert hooks, chains or straps in the rigging supports located at the top of the unit, between the fans (see drawing above).
2. Lift unit from the flat bed and move to a clear and completely horizontal area.

Note: Determine unit weight from the offer-related documents. The crane capacity, hooks, chains, straps and load bar must be determined accordingly.

Unpacking instructions

Once the unit has been placed over a completely horizontal area:

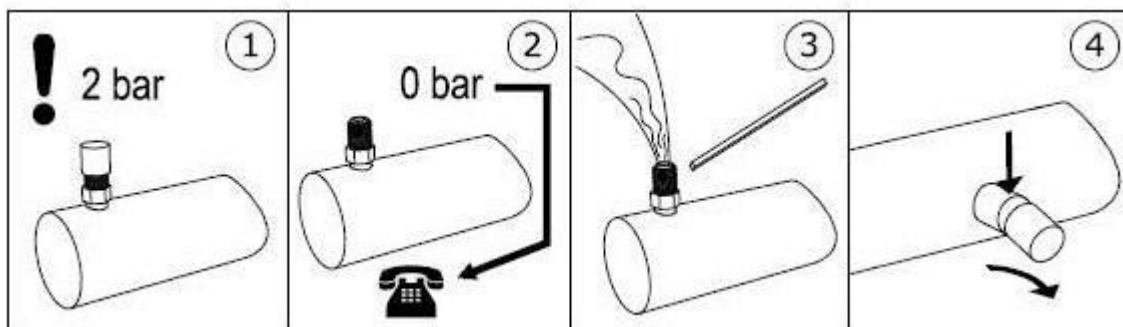
1. Remove plastic
2. Remove cardboard from corners
3. Remove wooden screws from the basement of the unit, that attach it to the pallet
4. Use crane and load bar to lift the unit and remove from the pallet
5. Place unit in site of operation
6. Set up installation of inlet and outlet pipelines and remove flange supports

Check transport excess pressure: The units are delivered by the manufacturer with approx. 2 bar transport excess pressure (cleaned and dried air). Check transport pressure at the Schrader

valve (pressure measurement). For pressure-less unit: Report immediately to manufacturer and note missing pressure on delivery note. An unpressurized unit indicates it presents leaks.

⚠ WARNING

Danger of injuries and damage to property due to escaping convector fluid! A pressure-less unit is sign of a leak on the unit due to transport damage. Escaping convector fluid due to leaks on the unit can lead to injuries (see Residual hazards with the heat carrier section, page 20). Do not take the unit into operation!



- 1 Transport pressure filling
- 2 Check transport pressure; inform manufacturer with unpressurized unit
- 3 Release transport pressure
- 4 Remove blanking plugs.

NOTICE

Danger of corrosion and dirt build-up!

Moisture and dirt must be prevented from entering the unit.

Protect the unit against dust, dirt, moisture, wet conditions, damaging and other detrimental influences. Detrimental influences (see Safety instructions for set-up and start-up section, page 29).

Begin with the installation as soon as possible.

8.4 Installation

8.4.1 System-side requirements for stress-free installation

Prevent stresses in the unit:

- ° Ensure that all fixing points have the same spacing to the fixing level.
- ° Ensure that all fixing points maintain the same spacing to the fixing level under load and permanently.

Set up and fix units as follows: Airflow must not be impaired by obstructions.

The units must be installed on fixing points that are appropriate for the unit's weight and then bolted with fixing bolts. The operator or installer of the equipment is responsible for ensuring that the bolted connections are of an adequate strength. The following instructions must be observed when fixing the units:

The diameter of the mounting holes has been statically determined by the manufacturer; the fixing bolts must be adapted accordingly. When calculating the transferring bearing strength it is imperative to take into account the total weight of the unit (= structural weight + weight of pipe content + additional weight, such as water, frost, ice, dirt or similar).

The fixing bolts must be secured against loosening with an appropriate locking device.

The fixing bolts must not be over-tightened or stripped.

All fixing bolts must be tightened equally.

Prevent the unit from shifting in its position. Fix the unit in its position. Tighten the fixing bolts and secure them against loosening.

Only fix the unit to the intended fixing points.

Güntner recommends the use of expansion joints in order to prevent stress in the connections.

8.4.2 Mounting the unit

⚠ WARNING

Danger of injuries with escaping heat carrier!

In case of improper installation, leak of heat carrier can occur during operation of the installation, this can lead to injuries or damage to property (see Residual hazards with the heat carrier section, page 20).

8.5 Notes on connecting the unit

WARNING

Danger of injuries and damage to property caused by escaping heat carrier!

In case of improper installation, leak of heat carrier can occur during operation of the installation, this can lead to injuries or damage to property (see Residual hazards with the heat carrier section, page 20).

Prevent heat carrier from escaping from the unit into the environment.

Secure all heat carrier-carrying lines against mechanical damage.

In areas that are used for internal traffic, only lay the pipelines to and from the unit with connections and fittings that cannot be removed.

Ensure that the on-site connections do not exercise any forces upon the distribution and header points. This can cause leaks on the heat carrier connection points of the unit and on connection points of the on-site pipe-laying.

Ensure that:

Release devices to prevent liquid escapes are provided and available.

8.5.1 Connect the unit to the heat carrier circuit of the system

WARNING

Danger of injuries and damage to property!

Improper connection to the heat carrier circuit of the system causes hazards:

Welding work on pressurized parts can result in fires or explosions.

Leaks result in escaping heat carrier (see Residual hazards with the heat carrier section, page 20).

Soldering and welding work on pressurized parts can result in fires or explosions.

Smoking or open light can cause fires or explosions.

Ensure that stresses and vibrations from the system including the heat carrier circuit are not passed on to the unit.

Only lay heat carrier-side connections stress-free! The on-site pipeline system must be braced before connecting to the unit!

Soldering and welding work is only permitted on unpressurized units!

Evacuate the unit professionally in acc. with current regulations.

The use of open fire at the installation site is forbidden. Fire extinguishers and extinguishing agents used to protect the equipment and the operating staff must comply with the requirements of current regulations.

Install the pipes in accordance with local regulations.

Ensure that:

Release devices to prevent liquid escapes must be provided and available.

That the pipeline installation is kept as short as possible. Use as few bends as possible, and if bends are necessary, use big radii!

The free space around the unit must be big enough to ensure there is no danger to the unit; regular maintenance of the components must be possible, and it must also be possible to check and repair components, pipes and fittings.

It must be possible to shut off the unit if a leak occurs. It must be possible to activate all devices used for diverting escaping heat carriers from a safe position.

Electrical components, e.g. for fan operation, for ventilation, for illumination and for alarm system on installation site, have to be designed with regard to condensation of air humidity and droplet formation in acc. with electrical local and national codes and regulations.

Observe the following when soldering:

- All connections must be hard-soldered!
- Avoid percussive soldering joints; use copper pipe ends that are widened on one side (capillary soldering)!
- Prevent leaks, solder precisely and carefully!
- Prevent overheating when soldering (danger of excessive scaling)!
- Use shielding gas when soldering (prevent excessive scaling)!

After completing pipe installation and before connecting the units they must be professionally cleaned internally and evacuated in accordance with local regulations.

8.5.2 Unit electrical connection and protection

All fans with rotating current drive can be operated via D-S switch-over with two different speeds:

- D: high speed
- S: lower speed

The fans can be speed-controlled with the use of Güntner control units.

Connect fan motors in acc. with motor connection circuit and check connection. Provide power supply in acc. with the specifications on the fan motor's type plate:

The fan motors are operated in 3~ rotating current (IP 54). Connect thermo contacts for motor protection.

Seal all electrical lines to the connection boxes in acc. with their protection class. The connection class is specified in the order-related offer documents.

NOTE! Danger of damaging! If the fusing is too high there is the danger of injuries to people and damage to property. The supply lines must always be fused in acc. with the smallest wire cross-section.

8.6 Perform acceptance test

WARNING

Danger of injuries and damage to property!

Escaping heat carrier can cause injuries or even death (see Residual hazards with the heat carrier section, page 20).

Perform the following acceptance test with an expert before starting up the unit after making important changes and after a unit swap-out.

Ensure that the temperature at the set-up point corresponds with the technical data (see Technical data section, page 25).
Ensure that sufficient air can be sucked in and released.

Ensure that the supply feed is sufficient for the required energy. Compare the unit inside the heat carrier circuit of the system with the plans of the system including the heat carrier circuit and the electrical wiring diagrams.

Test the unit for vibrations and movements that can be caused by the fans and operation of the system including the heat carrier circuit. Remove oscillations, vibrations and movements following consultation with the manufacturer or independently. Perform visual inspection of the structural design, brackets and fixtures (materials, sequence, connections) the ability to operate and the arrangement of fittings.

Check and re-tighten all threaded connections, especially on the fans.

Check installation of pipe connections.

Check laying of heat carrier-carrying connection pipes. Ensure that the unit is protected against mechanical damage.

Ensure that the unit is protected against inadmissible heating up and cooling down.

Check the fan blades protection.

Ensure that optimum unit control and accessibility are guaranteed.

- Is the unit placed so that it can be monitored and controlled from all sides at all times?
- Is sufficient space provided for maintenance?
- Are all liquid-carrying components, connections and lines and all electrical connections and lines easy to access?
- Are the pipes easy to identify?

Check heat exchanger surfaces for dirt and clean as required (see Clean unit page 49). Perform function tests on the fans (rotation direction, power consumption, etc.).

Check electrical connections of the fan motors for damage.

Check the quality of the soldering connections, the electrical connections and the fixture connections.

Perform pressure test with test gas and with a test pressure of 1.1-times the permissible operating pressure: check the connection seals and detect leaks, e.g. with foaming agent, or similar.

Check corrosion protection: Perform a visual inspection on all bends, components and component holders that are not heat-insulated. Document and archive test results.

Perform test run. Observe and check unit during test run, in particular for:

- Fan smooth running (bearing noises, contact noises, imbalances, etc.).
- Fan power consumption:
- Leaks

Report all defects to the manufacturer immediately. Remove defects following consultation with the manufacturer. Check the unit and the unit's interaction with the heat carrier circuit of the system again after 48 operating hours, especially on the connections and on the fans, and document the test results.

8.7 Test readiness for operation

Ensure that all electrical protective measures are ready to function. Ensure that all heat carrier-carrying connections are securely in place. Ensure that all electrical connections (fans) are effected securely.

8.8 Operating the unit for the first time

WARNING

Danger of injuries and damage to property!

Escaping heat carrier can cause injuries or even death (see Residual hazards with the heat carrier section, page 20).

Only put the unit into operation, when:

- The units have been properly mounted and connected (see Installation section, page 37).
- You have performed a complete acceptance test (see Perform acceptance test section, page 40).
- You have performed a test for ensuring the system's readiness for operation (see Test readiness for operation section, page 41) and all safety precautions (see Safety section, page 11) have been taken. Follow the operating instructions manual of the system!

Contact the manufacturer immediately if you want to operate the unit under different operating conditions as those defined in the order-related offer documents.

Switch on the system including the heat carrier circuit and of the electrical system Activate the unit:

Open the valves on the inlet and outlet-side heat carrier circuit of the system.

Activate fans

Wait until the operating point is reached. After the operating point is reached the unit is ready to operate.

Parameters for setting the operating point: see order-related offer documents.

Operating point:

- Heat carrier flow temperature
- Heat carrier return temperature or circulating heat carrier volume
- Airflow volume
- Air inlet temperature

To ensure the specified operating point is complied with the actuators for the operating point setting must be secured against unauthorized access (e.g. by sealing, screwing on caps, removing hand wheels, etc.).

9 Operation

9.1 Safety

⚠ WARNING	
	Danger of cutting off, pulling in! There is a danger of cutting off fingers on the rotating fan blades, injury hazard for the hands and pulling in danger with loose elements such as hair, necklaces or clothing parts.
⚠ WARNING	
	Burn hazard! Danger of burns if parts are touched (Thermal residual hazards section, page 19). Do not touch any parts of the unit without protective gloves when the unit is in operation or if it has not yet cooled down again after operation.fans without guard grille!

9.2 Operating the unit

To operate the unit, the system must be in operation, including the heat carrier circuit and the electrical system. The unit must be started up by opening the respective valves on the inlet and outlet-side of the heat carrier circuit and by connecting to the electrical system

- Switch on electrical system
- Open heat carrier-carrying lines
- Switch on fans

9.3 Taking the unit out of operation

The units are system components of a refrigeration system, including the heat carrier circuit. The unit is taken out of operation by switching off the heat carrier circuit of the system in acc. with the system operating instructions manual. To do this, the heat carrier-carrying lines must be shut off from the system and the fans must be disconnected from the electrical system:

- Switch fans off
- Switch electrical system off
- Close heat carrier-carrying lines

NOTE! When shut down, consider the maximum operating pressure! If necessary, take precautions so that it cannot be exceeded.

NOTICE

With shutdown times of a month or longer put the fans into operation for approx. 2-4 hours a month to maintain their functionality.

9.4 Shutting the unit down

⚠ WARNING

Danger of injuries and damage to property!

Escaping heat carrier can cause injuries (see Residual hazards with the heat carrier section, page 20), and see Residual hazards caused by pressurized parts section, page 21).

Ensure that the maximum operating pressure is not exceeded after the shutdown either!

NOTICE

Danger of corrosion and dirt build-up!

Moisture and dirt must be prevented from entering the unit.

Protect the unit against dust, dirt, moisture, wet conditions, damaging and other detrimental influences (see Safety instructions for set-up and start-up section, page 29).

With shutdown times of a month or longer put the fans into operation for approx. 2-4 hours a month to maintain their functionality.

Take the unit out of operation (see Taking the unit out of operation section, page 42). Secure the unit:

When shutdown consider max. operating pressure! (see Technical data section, page 25). If necessary, take precautions so that it cannot be exceeded.

Secure the fan motor drives against switching on again

Secure heat carrier-carrying lines against pressurizing with heat carrier.

Secure unit against detrimental influences at the installation or storage location (see Safety instructions for set-up and start-up section, page 29) in order to keep all unit components in good state for proper use and to conserve the usability of the unit. For this purpose, special storage conditions (see Storage before installation, page 29) and preventive measures for corrosion protection have to be provided, regular testing of the functionality of the fans and of the unit taken out of operation have to be effected.

Drain the unit: Completely release the heat carrier (see Residual hazards with disposal section, page 24).

9.5 Operating the unit after a shutdown

The unit must be put back into operation after a shutdown in acc. with the system-specific set-up in acc. with the operating instructions manual as follows:

Test the unit's readiness for operation (see Test readiness for operation section, page 41). Perform pressure test and visual inspection for corrosion protection.

NOTE! The pressure test with commissioning a second time must only be carried out with appropriate media at appropriate test pressure.

Put the unit into operation (see Operating the unit section, page 41).

9.6 Changing the unit over to another heat carrier

WARNING

Danger of injuries and damage to property!

Considerable dangers can be caused with operation with another heat carrier without prior manufacturer approval (Improper use).

The unit must not be changed over to another heat carrier without prior written consent by Güntner.

- Ensure that the unit manufacturer has agreed to the changeover.
- Ensure that the right heat carrier is re-filled. Ensure that all materials used in the unit are compatible with the new heat carrier.
- Ensure that the permissible pressure is not exceeded.
- Check that the new heat carrier can be used without requiring a new test certificate for the unit. Ensure that classification is complied with.
- The safety device for the unit must be swapped out or reset.
- Mixtures with residues of heat carrier, for example, must be prevented.
- All information concerning the new heat carrier must be changed accordingly.
- The complete documentation, including these operating instructions and the system operating instructions manual must be changed accordingly.
- Perform acceptance test (see Perform acceptance test section, page 40).

9.7 Water quality parameters for copper tubes

The following water quality parameters are recommended to be used during the operation and inspection of the Güntner US, LLC (Güntner) family of finned copper tube heat exchange products designed and fabricated by Güntner.

Following these guidelines will help maximize the lifetime of the equipment and to minimize the risk of failure in service. All recommendations made in this guide are brand-independent, and it is up to each operator to select the most convenient chemical supplier, monitoring technology, etc. for their particular application and location.

Correct control of the water quality of the circulating fluid will help to prevent or mitigate a whole host of detrimental mechanisms that can lead to failure of the unit.

Failure to follow recommendations or warnings listed in the document may result in damage to the unit and subsequent loss of service. Güntner is not liable for failures induced by the operator choosing to ignore the recommendations in this section.

Water quality parameters during operation for copper tubes

Parameter	Value	Units
pH	7.5 - 9.0	
Specific Conductance, 25° C	337	µS
Alkalinity as CaCO ₃ — "M"	119	mg/l
Alkalinity as CaCO ₃ — "P"	0	mg/l
Aluminum as Al	0.25	mg/l
Arsenic as As	<0.003	mg/l
Barium as Ba	0.02	mg/l
Cadmium	<0.005	mg/l
Calcium as CaCO ₃	87.4	mg/l
Chloride as Cl	30.8	mg/l
Chromium	<0.005	mg/l
Copper as Cu	<0.005	mg/l
Fluoride as F	1.0	mg/l
Iron as Fe	0.1	mg/l
Lead as Pb	<0.005	mg/l
Magnesium as CaCO ₃	48.6	mg/l
Manganese as Mn	<0.005	mg/l
Nitrate as NO ₃	0.6	mg/l
Oxygen	< 1	mg/l
Phosphate as PO ₄	<0.05	mg/l
Potassium as K	ND	mg/l
Silica as SiO ₂	1.7	mg/l
Sodium as Na	11.0	mg/l
Strontium as Sr	ND	mg/l
Sulfur as SO ₄	10.9	mg/l
Zinc as Zn	0.03	mg/l
Color	<5	CU
Turbidity	0.1	NTU
Total Dissolved Solid	188	mg/l
Total hardness	ppm or mg/l of CaCO ₃	< 534
Total bacteria	CFU/ml	< 1000

NOTE:

- 1) The airtightness of the circuit must be guaranteed at all times. Air access leads to sharp pH fluctuations and a corrosion damage to the copper tubes.
- 2) When the circuit is opened, the inflow of air must be prevented using protective atmospheres.
- 3) It is recommended a constant supervision of the pH of the circulating water.
- 4) It is highly recommended to keep a professionally competent surveillance of the operation of the system (tightness of the system, control of the pH values, controls and calibration of measuring equipment, maintenance and repairs).
- 5) The responsibility for compliance with the limit values lies with the system manufacturer or operator.

10 Troubleshooting

10.1 Safety

WARNING

Danger of injuries and damage to property!

Faults that are not described in these operating instructions must only be removed by Güntner. Contact the Güntner hotline.

Faults that are described in these operating instructions must only be removed by appropriately trained personnel (see Personnel, care requirements section, page 15).

With faults that occur during the operation, monitoring and maintenance of the complete system, inform Güntner immediately.

10.2 Service

Office hours

Tel. +1 847 781 0900

Fax. +1 847 781 0901

serviceus@guntnerus.com

www.guntnerus.com

10.3 Troubleshooting table

Fault	Possible cause(s)	Remedy
Fan motor not running	Power supply interrupted	Restore power supply
	Fan blade stuck	Enable fan to rotate freely
Bearing noises	Defective fan motor	Renew bearing or fan motor
Unit vibrating	Defective fan blades	Change fan blades
	Loose fan fixture	Tighten fixtures
Unit capacity not achieved	Coil is strongly contaminated with dirt	Clean
	Fans not running properly or down	Repair, swap out fans
	Heat carrier pressurizing defective (temperature and quantity insufficient)	Set heat carrier pressurizing values (temperature and quantity) to reference values.
Unit capacity not achieved	Insufficient heat exchanger setting	Set heat exchanger setting (con- centration, inhibition)
Heat carrier escaping	Unit heat carrier-carrying components leaking	Switch off heat carrier feed and fans, close leak

11 Maintenance

11.1 Safety

11.1.1 Before starting all maintenance

⚠ WARNING

Danger of injuries and damage to property with escaping heat carrier! (see Residual hazards with the heat carrier section, page 20).

Only perform maintenance work – especially welding work – on the leaking unit after completely removing the heat carrier from the leaking unit!

Perform the following safety measures before beginning all maintenance work:

Evacuate the unit's heat exchanger

Clean and blow out the unit's heat exchanger.

11.1.2 With all maintenance work

⚠ WARNING

Danger of injuries and damage to property with escaping heat carrier! (see Residual hazards with the heat carrier section, page 20).

Escaping heat carrier with leaks on the unit can cause the following hazard situations and injuries:

	Warns against hot surfaces! In operation the unit's heat exchanger and pipes have temperatures of over +45°C. Contact can cause burns. With heat carrier temperatures over +45 °C there is the hazard of burns on pipe components and distribution pipes.
	Use hand protection!
	Warns against freezing! With insufficient frost protection filling and operation in areas with frost hazard the unit can cause freezing. With units that cannot be drained completely the hazard of freezing also remains after draining. It is imperative that the proper ventilation be ensured when draining the unit. With pressure test, operation and shutdown of units filled with water or insufficient frost protection filling or insufficient setting of the heat carrier, these are disrupted at minus temperatures.
	• Use eye protection!
	• Use eye protection!

• Ensure that the unit in question is pressure-free before maintenance work begins or the working fluid is sucked up from the unit in question.
 • Power off the electrical system and secure it against unintentional switching on again.
 • Separate the unit to be maintained from the heat carrier circuit of the system and secure it.

NOTICE



Danger of damage to property!

With work in the inlet and outlet feeds of the fans and heat exchanger, objects can get into the fans and therefore cause faults and damage on the components.

- Power off the fans before beginning maintenance work and secure them against switching on again.
- After the work has finished do not allow any objects to get back into the inlet and outlet feeds of the fans or the set-up room.

⚠ WARNING

Only step on to the units when required and only with firm, tough and safe footwear. If the units do not have any railings, always only walk on them when required and with a safety device that prevents falling.

11.1.3 After all maintenance work

⚠ WARNING

Danger of injuries and damage to property caused by escaping heat carrier (see Residual hazards with the heat carrier section, page 20).

Perform the following safety measures after finishing all maintenance work:

- Ensure the switching and activation devices, the measuring and display devices and the safety devices function properly.
- Ensure the heat carrier fittings are functioning.
- Check the identification of the pipelines and ensure this is visible and legible.
- Check the fixing and corrosion protection of the components in question.
- Ensure the electrical connections (fans) are functioning.
- Perform a pressure test and a tightness test.
- Perform an acceptance test (see Perform acceptance test section, page 40).
- Perform a pressure test.

11.2 Inspection and maintenance plan

The checks to be performed are listed in the following sections as time-scaled checklists.

11.2.1 Fans

The manufacturer's operating specifications take priority for this component. Güntner recommends proceeding in acc. with the following inspection and maintenance plan:

d = daily, w = weekly, m = monthly, y = yearly				
Work to be performed	d	w	m	y
Check the fans for dirt build-ups. • With dirt build-ups: Clean fans (see Clean unit section, page 49)	X			
Check the fan drive for smooth running. • Vibrations on the unit: remove imbalances • Where required, tighten and correct blade fixtures and blade settings		X		
Fan bearing: Running sound and smoothness changes • Swap out bearing			X	
Fan motor: Is a new bearing due? • Swap out bearing or motor; where applicable, clean and repair motor				X
Fan wheel: corrosion on bolts (with threaded blades) • Swap out bolts				X
Fan blades: Corrosion or damage on blades • Swap out blades or wheel				X

(*) Recommended: every six months

11.2.2 Unit heat exchanger

⚠ WARNING

Danger of injuries and damage to property caused by escaping heat carrier! (see Residual hazards with the heat carrier section, page 20)

d = daily, w = weekly, m = monthly, y = yearly				
Work to be performed	d	w	m	y
Check heat exchanger for dirt build-ups. • With dirt build-ups: Clean heat exchanger (see Clean unit section, page 49)	X			

d = daily, w = weekly, m = monthly, y = yearly				
Work to be performed	d	w	m	y
Check heat exchanger's general condition. • Damages found: remove damages		X		
Check heat exchanger's operating point (see Operation section, page 42). • Change in fan capacity found: restore required system-side conditions (including heat carrier circuit). • Change in surface temperatures found: restore required system-side conditions (including heat carrier circuit).			X	
Change in heat exchanger setting found: restore required heat exchanger setting (concentration, inhibition)				X
Check heat exchanger and connections for tightness. • Repair affected unit part (see Remove leaks section, page 49).				X
Check heat exchanger for heat carrier pressurizing. • Restore required heat carrier-side conditions.				X
Check heat exchanger for corrosion. • Corrosion or damage on core tubes, fins, support structures, tube connections, fixtures: repair unit sections affected.				X

(*) recommended: every six months

Care and maintenance plan

Measure	Cleaning agent	Interval
Component cleaning	Mechanically	As required (visual inspection)
Complete cleaning	Warm water or environmentally safe cleaning agent	After 120 hours
Leak check		After 6 months
Heat exchanger setting check		After 6 months
Corrosion protection check		After 6 months

11.3 Maintenance work

11.3.1 Remove leaks

⚠ WARNING

Danger of injuries with heat carrier! (see Residual hazards with the heat carrier section, page 20).

- Have leaks removed as quickly as possible by an expert.
- Do not fill with a heat carrier other than that specified in acc. with the order-related offer documents!
- Only put the unit back into operation when all leaks have been repaired.

Perform all work including pressure, acceptance and functional test (see Perform acceptance test section, page 40, and see Test readiness for operation section, page 41).

11.4 Clean unit

11.4.1 General

The following applies for cleaning: The operator must ensure that the cleaning agent is environmental friendly. Harmful to the environment substances, e.g. acid-forming, are forbidden.

Clean casing by rinsing with warm water (approx. +25°C) and/or with environmentally friendly cleaning agents.

Rinse thoroughly with water after using cleaning agents.

Allow casing to dry completely.

Check heat carrier-carrying and electrical connections (see Test readiness for operation section, page 35).

11.4.2 Clean heat exchanger

Drain the unit. Shut off the unit. Power off the fans. Perform defrosting.

Clean the heat exchanger with one of the following procedures:

Cleaning with compressed air (see Cleaning with compressed air section, page 53).

Hydraulic cleaning (see Hydraulic cleaning section, page 50)

Cleaning with brushes or brushes and compressed air (see Cleaning with brushes and compressed air section, page 53).

NOTICE

Danger of damage to property!

The fins can be damaged with too high a pressure, too small a distance or a cleaning jet that hits the fins at an angle. Mechanical cleaning with hard objects (e.g. steel brushes, screw drivers, or similar) damages the heat exchanger.

- Use a pressure of max. 50 bar with hydraulic cleaning and max. 80 bar with cleaning with compressed air!
- Maintain a minimum distance from the fins of 200 mm!
- Always aim the jet vertically (max. ± 5 degree deviation) at the fins!
- Do not use any hard objects when cleaning!

- Switch fans on.
- Open drain side.
- Open inlet side.

11.4.2.1 Cleaning with compressed air

Blast heat exchanger (max. 80 bar pressure) with compressed air to remove dirt and contaminants.

NOTE! Hold the jet of the cleaner vertical to the fins (max. deviation of ± 5 degrees) to prevent fins from bending.

10.4.2.2 Hydraulic cleaning

⚠ WARNING



Warns against dangerous electrical voltage!

Direct and indirect contact with voltage-carrying parts of motors and electrical lines can cause serious injuries or death. Water and cleaning agents conduct electricity.

- Power off the fans for cleaning with water or steam jet and ensure them against switching on again.

NOTICE

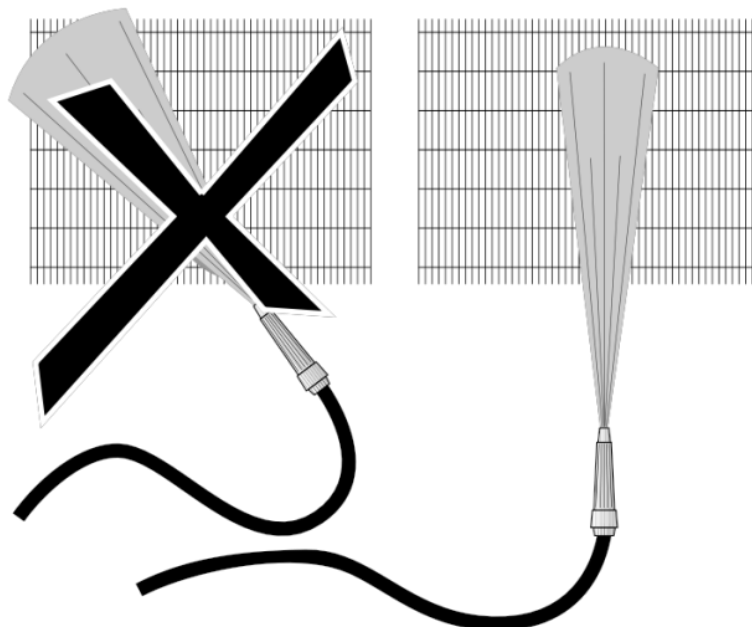
Danger of damage to property!

Water or steam jets can damage fans, electrical lines or other components.

- Ensure that electrical connections and motors, as well as components and stored goods at the set-up point are not touched by water or steam jets. Cover these if required.

Heavier moisture or greasy dirt must be removed with a high-pressure water jet (max. 50 bar pressure), steam pressure jet (max. 50 bar pressure), min. 200 mm distance with flat jet nozzle, or using neutral cleaning agent where applicable, against the coil. Ensure here:

- With oily and greasy dirt it helps to add a cleaning agent to the water.
- When applying chemical agents ensure that they are not aggressive towards the unit's materials. Rinse the unit off after the treatment. The use of chemical substances is always under the full responsibility of the enduser and/or customer.
- Cleaning should be done from one side of coil preferably from outside to the inside, that is enough if done frequently. When the coil is too dirty and/or it has been reduced significantly its capacity, it must be cleaned from both sides.
- Hold the jet of the cleaner vertical to the fins (heat exchanger – max. deviation of ± 5 degrees) to prevent fins from bending



- Continue cleaning until all dirt has been removed.

11.4.2.3 Cleaning with brushes and compressed air

Remove dry dust or dirt with a brush, a hand brush or with compressed air (max. 80 bar pressure, min. distance of 200 mm to the fins; against the airflow direction) or with a powerful industrial vacuum cleaner.

- Use soft brushes! (no steel brushes or similar)
- Where possible, cleaning should be from the top down, so that the dirt can be removed without the possibility of returning to cleaned parts. Continue cleaning until all dirt has been removed.
- Always brush the fins lengthways – never across!

11.4.2.4 Cleaning the pipe insides

Rinsing with solvents removes dirt build-ups formed by lime scale, paint-type layers, oil residues, etc. For cleaning purpose please consult the chemical compatibility between the solvent and the core tube.

Cleaning principle:

- 1 Pump
- 2 Heat exchanger
- 3 Containers for catching cleaning liquid

Chemical cleaning is applied where the dirt is stubborn or with heat exchangers with lime scaled pipes and collectors. Select the pump's supply rate so that a speed that is adjusted to the cleaning effect is achieved (2.5 to 5 m/s).

Neutralize and rinse the heat exchanger well after cleaning.

WARNING

Toxic hazard!

Trichlorethylene is harmful to the health and highly narcotic: inhalation of air with 200 ppm causes tiredness. It has been classified by the MAC Commission as carcinogenic (category 2) and germ cell mutagenic (category 3B). It must be identified as "Toxic".

Acute poisoning can cause brain damage, blindness and destruction of the smell and taste senses.

- Only use trichlorethylene in closed systems with complete recovery.
- Exclude all open flames or spark sources.
- Smoking is forbidden.
- Protect the skin and eyes with suitable protective materials.

11.4.3 Cleaning fans

WARNING

Danger of cutting off, pulling in!

There is a danger of cutting off fingers on the rotating fan blades, injury hazard for the hands and pulling in danger for loose elements such as hair, necklaces or clothing parts.

•Power off the unit before you begin maintenance work. Secure the unit against unintentional switching on again by removing the electric fuses for the unit. Secure the unit with a suitable warning sign referring to unintentional switching on.

•Put fans and guard grilles that were removed or opened for maintenance back into their original status without fail before you put the unit back into operation!

Dirt build-ups on the fans, the streamers or air outlet connectors (if installed) and the fan guard grilles must be removed regularly, otherwise they will cause imbalances or destruction and power losses. The fan motors themselves are maintenance-free.

Power off the unit and secure it against unintentional switching on.

Clean fans with one of the following procedures:

NOTE! Danger of damage to property! Mechanical cleaning with hard objects (e.g. steel brushes, screw drivers, or similar) damages the fan. Forbidden!

- Cleaning with compressed air: Blow fans with compressed air (max. 10 bar pressure; min. distance 200 mm) to remove dirt and contaminants. Continue cleaning until all dirt has been removed.
- Cleaning with compressed air and brushes: Remove dry dust or dirt with a brush, a hand brush or with compressed air (max. 10 bar pressure, min. distance of 200 mm) or with a powerful industrial vacuum cleaner. Ensure here: use soft brushes! (no steel brushes or similar). Continue cleaning until all dirt has been removed.

Mount guard grille, streamer or air inlet connections (if installed)

- Switch unit on

12 Plans/diagrams

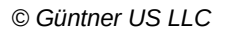
12.1 Electrics documentation

12.1.1 Fan motor connection diagram

See motor terminal box cover inside on the fan.

12.2 Heat carrier connection diagram

See order-related offer documents



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